

C215

Operations Management

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A medical-supply distributor with a new warehouse in Reno sells 18,000 boxes of gloves a year. Each order costs \$40 to place, and holding one box for a year costs \$4. The supplier offers free shipping on orders over 1,000 boxes. What is the economic order quantity?

- A. 1,000 boxes
- B. 600 boxes
- C. 60 boxes
- D. 424 boxes

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A firm's annual demand is 18,000 units and its EOQ is 600 units. How many orders will it place per year?

- A. 600
- B. 30
- C. 0.033
- D. 300

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A hardware store sells 40 bags of concrete a day. The supplier, a family business with an excellent reputation, delivers 5 days after an order, and the store keeps 60 bags of safety stock. Each order costs \$75 to process. At what inventory level should it reorder?

- A. 60 bags
- B. 200 bags
- C. 140 bags
- D. 260 bags

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A clinic reorders syringes at 500 units. Daily use is 50 units and lead time is 8 days. How much safety stock is it carrying?

- A.** 100 units
- B.** 900 units
- C.** 10 units
- D.** 400 units

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

Store-level sales of a popular sports drink vary about 5% week to week. Yet the bottler, which recently upgraded its bottling line and sponsors a local marathon, sees weekly orders from distributors swing by 40%. Each distributor forecasts on its own and orders in large batches. What best explains the pattern?

- A.** The bullwhip effect, driven by independent forecasting and batch ordering.
- B.** Consumer demand is highly unstable.
- C.** Each stage should hold much larger buffers to absorb it.
- D.** The new bottling line is unreliable.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A bakery's ovens have a design capacity of 1,000 loaves a day. After cleaning and changeovers, effective capacity is 800 loaves. Yesterday the bakery, which was featured on a local news segment, produced 720 loaves. What was its utilization?

- A.** 72%
- B.** 138.9%
- C.** 80%
- D.** 90%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A line produced 540 units with an efficiency of 90%. What is its effective capacity?

- A.** 486 units
- B.** It cannot be found without design capacity.
- C.** 540 units
- D.** 600 units

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

An order-fulfillment process has three steps: picking handles 60 orders an hour, packing 45 an hour, and labeling 50 an hour. Picking uses a new robotic system that cost \$2 million. What is the maximum output of the process?

- A.** 60 orders an hour
- B.** 155 orders an hour
- C.** 45 orders an hour
- D.** 51.7 orders an hour

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A loan office processes 20 applications an hour, and each application spends an average of 3 hours in the system. The office opened a second location last spring and uses a cloud-based workflow tool. On average, how many applications are in process at any time?

- A.** 23
- B.** 6.7
- C.** 20
- D.** 60

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A repair shop averages 120 jobs in process and completes 40 jobs a day. How long does a job spend in the shop on average?

- A.** 4,800 days
- B.** 3 days
- C.** 40 days
- D.** 0.33 days

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A print shop runs one 8-hour shift with 4 workers and produces 480 brochures. It recently switched to recycled paper and raised prices 3%. What is labor productivity?

- A.** 0.067 labor-hours per brochure
- B.** 60 brochures per hour
- C.** 15 brochures per labor-hour
- D.** 120 brochures per worker

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A café chain with 12 locations forecast 200 cases of oat milk for last week and actually used 240. It uses exponential smoothing with $\alpha = 0.3$, and its regional manager expects a warm spring. What is the forecast for this week?

- A.** 12 cases
- B.** 240 cases
- C.** 212 cases
- D.** 228 cases

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A medical-supply distributor ~~with a new warehouse in Reno~~ sells 18,000 boxes of gloves a year. Each order costs \$40 to place, and holding one box for a year costs \$4. ~~The supplier offers free shipping on orders over 1,000 boxes.~~ What is the economic order quantity?

Target word: EOQ in boxes **Decides:** $\sqrt{2 \times 18,000 \times 40 / 4}$.

A	Camouflage	The free-shipping threshold, which is not part of the EOQ formula.
B	Correct	$2 \times 18,000 \times 40 / 4 = 360,000$; $\sqrt{360,000} = 600$.
C	Inverted operation	Swaps S and H: $\sqrt{2 \times 18,000 \times 4 / 40} = 60$.
D	Dropped element	Drops the 2: $\sqrt{(18,000 \times 40 / 4)} = \sqrt{180,000} \approx 424$.

2. Answer B Concept 1 • transfer

A firm's annual demand is 18,000 units and its EOQ is 600 units. How many orders will it place per year?

Target word: orders per year **Decides:** D / Q .

A	Wrong stem number	The order size, not the count.
B	Correct	$18,000 / 600 = 30$ orders.
C	Inverted operation	$600 / 18,000$ inverts the ratio.
D	Adjacent category	That is average inventory ($Q/2$), not number of orders.

3. Answer D Concept 2

A hardware store sells 40 bags of concrete a day. The supplier, ~~a family business with an excellent reputation,~~ delivers 5 days after an order, and the store keeps 60 bags of safety stock. ~~Each order costs \$75 to process.~~ At what inventory level should it reorder?

Target word: reorder point **Decides:** $40 \times 5 + 60$.

A	Wrong stem number	The safety stock alone.
B	Dropped element	Leaves out the safety stock.
C	Inverted operation	Subtracts safety stock instead of adding it.
D	Correct	Lead-time demand $40 \times 5 = 200$, plus 60 safety stock = 260.

4. Answer A Concept 2 • transfer

A clinic reorders syringes at 500 units. Daily use is 50 units and lead time is 8 days. How much safety stock is it carrying?

Target word: solve for safety stock **Decides:** $ROP - (d \times L)$.

A	Correct	$500 - (50 \times 8 = 400) = 100$.
B	Inverted operation	Adds lead-time demand to the reorder point instead of subtracting.
C	Wrong stem number	$500 / 50$ is days of supply, not safety stock.
D	Adjacent category	That is lead-time demand, not the buffer.

5. Answer A Concept 3

Store-level sales of a popular sports drink vary about 5% week to week. Yet the bottler, ~~which recently upgraded its bottling line and sponsors a local marathon,~~ sees weekly orders from distributors swing by 40%. Each distributor forecasts on its own and orders in large batches. What best explains the pattern?

Target word: name the cause **Decides:** Variability grows upstream from independent forecasts and batching.

A	Correct	Each stage amplifies demand swings; shared sales data and smaller orders reduce it.
B	Inverted operation	Store sales vary only 5%; the swings are created upstream.
C	Adjacent category	Bigger buffers treat the symptom and often feed the effect.
D	Camouflage	The upgrade does not explain distributor order swings.

6. Answer A Concept 4

A bakery's ovens have a design capacity of 1,000 loaves a day. After cleaning and changeovers, effective capacity is 800 loaves. Yesterday the bakery, ~~which was featured on a local news segment,~~ produced 720 loaves. What was its utilization?

Target word: utilization **Decides:** Actual / design.

A	Correct	$720 / 1,000 = 72\%$.
B	Inverted operation	$1,000 / 720$ inverts the ratio.
C	Adjacent category	$800 / 1,000$ compares effective to design capacity.
D	Confusion set	$720 / 800$ is efficiency, not utilization.

7. Answer D Concept 4 • transfer

A line produced 540 units with an efficiency of 90%. What is its effective capacity?

Target word: solve for effective capacity **Decides:** Effective = actual / efficiency.

A	Inverted operation	Multiplies by 0.90 instead of dividing.
B	Adjacent category	Design capacity is needed for utilization, not efficiency.
C	Wrong stem number	The actual output, not the capacity.
D	Correct	$540 / 0.90 = 600$.

8. Answer C Concept 5

An order-fulfillment process has three steps: picking handles 60 orders an hour, packing 45 an hour, and labeling 50 an hour. ~~Picking uses a new robotic system that cost \$2 million.~~ What is the maximum output of the process?

Target word: process throughput **Decides:** The slowest step.

A	Camouflage	The robotic picking step's rate; it is not the constraint.
B	Inverted operation	Adds step rates; steps in sequence do not add.
C	Correct	Packing is the bottleneck; the process cannot outrun it.
D	Adjacent category	Averages the steps; throughput is set by the minimum.

9. Answer D Concept 6

A loan office processes 20 applications an hour, and each application spends an average of 3 hours in the system. ~~The office opened a second location last spring and uses a cloud-based workflow tool.~~ On average, how many applications are in process at any time?

Target word: WIP **Decides:** 20×3 .

A	Adjacent category	Adds rate and time; Little's Law multiplies them.
B	Inverted operation	Divides 20 by 3 instead of multiplying.
C	Wrong stem number	The hourly rate, not the number in process.
D	Correct	Inventory = throughput $\times$ flow time = $20 \times 3 = 60$.

10. Answer B Concept 6 · transfer

A repair shop averages 120 jobs in process and completes 40 jobs a day. How long does a job spend in the shop on average?

Target word: solve for flow time **Decides:** Flow time = WIP / throughput.

A	Inverted operation	Multiplies WIP by throughput.
B	Correct	$120 / 40 = 3$ days.
C	Wrong stem number	The daily completion rate, not the time.
D	Inverted operation	Divides throughput by WIP — the ratio upside down.

11. Answer C Concept 7

A print shop runs one 8-hour shift with 4 workers and produces 480 brochures. ~~It recently switched to recycled paper and raised prices 3%.~~ What is labor productivity?

Target word: units per labor-hour **Decides:** $480 / (4 \times 8)$.

A	Inverted operation	Inputs over outputs — the ratio upside down.
B	Dropped element	Divides by shift hours only, dropping the number of workers.
C	Correct	$4 \times 8 = 32$ labor-hours; $480 / 32 = 15$.
D	Adjacent category	Output per worker per shift, not per labor-hour.

12. Answer C Concept 8

A café chain ~~with 12 locations~~ forecast 200 cases of oat milk for last week and actually used 240. It uses exponential smoothing with $\alpha = 0.3$, ~~and its regional manager expects a warm spring.~~ What is the forecast for this week?

Target word: next forecast **Decides:** $200 + 0.3 \times (240 - 200)$.

A	Dropped element	Just the adjustment, dropping the prior forecast.
B	Wrong stem number	Last week's actual — a naive forecast, not smoothing.
C	Correct	$200 + 0.3 \times 40 = 212$.
D	Inverted operation	Uses 0.7 instead of α : $200 + 0.7 \times 40$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	7	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	13	
Absolute language	Always, never, guarantee — where judgment applies.	0	
Confusion set	Every option is real; the test is exact role-matching.	1	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Economic order quantity	$EOQ = \sqrt{(2DS / H)}$, where D is annual demand, S is cost per order, and H is annual holding cost per unit; it balances ordering cost against holding cost.	Dropping the 2, or swapping S and H.	Annual demand, an ordering cost, and a holding cost in one stem.
2	Reorder point	Reorder point = daily demand $\times$ lead time (days) + safety stock.	Leaving out safety stock or multiplying by the order quantity.	Daily demand, a lead time, and a buffer.
3	Bullwhip effect	Small swings in end-customer demand grow larger upstream as each stage forecasts and batches orders on its own; sharing point-of-sale data and smaller, more frequent orders dampen it.	Blaming the swings on end customers and responding with bigger buffers at each stage.	Order variability that grows moving from retailer to wholesaler to manufacturer.
4	Utilization vs. efficiency	Utilization = actual output / design capacity; efficiency = actual output / effective capacity.	Dividing by the wrong capacity figure.	Three capacity figures: design, effective, and actual.
5	The bottleneck sets throughput	A process can produce no faster than its slowest step; improving any non-bottleneck step does not raise output.	Averaging step capacities or improving the fastest step.	A sequence of steps each with its own rate.
6	Little's Law	Inventory (WIP) = throughput rate $\times$ flow time, in consistent time units.	Dividing when you should multiply, or mixing hours and days.	Two of: average units in the system, rate per period, time in the system.
7	Productivity	Productivity = outputs / inputs; labor productivity is units per labor-hour.	Dividing by workers or shift length alone, or inverting the ratio.	Output and a labor or machine input over a period.
8	Exponential smoothing	Next forecast = last forecast + $\alpha \times$ (last actual – last forecast).	Applying α to the forecast instead of the error, or using $(1 - \alpha)$.	A prior forecast, an actual, and a smoothing constant.

Course scope (public description): This course focuses on the strategic importance of operations management to overall performance. This course also emphasizes principles of supply chain management relevant to a variety of business operations ranging from manufacturing goods to retail services. You will examine the various planning, control, and decision-making tools and techniques of the operations function.